Diachronic NLP Analysis of “Therapy-Speak” in General Web Discourse: ADHD and Autism in Common Crawl

Jakob Lütkemeier

MSc Dissertation submitted in partial fulfilment of the requirements for the degree of

MSc in Applied Social Data Science

School of Social Sciences and Philosophy

Department of Political Science

Supervisor: Dr Tom Paskhalis

Course Directors: Dr Jeffrey Ziegler; Dr Tom Paskhalis

Trinity College Dublin

10 August 2026

Note: Student ID is intentionally not printed (university guidance).

Declaration

I hereby declare that this MSc Dissertation is entirely my own work and that it has not been submitted as an exercise for a degree at this or any other university.

I have read and I understand the plagiarism provisions in the General Regulations of the University Calendar for the current year, found at http://www.tcd.ie/calendar.

I have also completed the Online Tutorial on avoiding plagiarism "Ready Steady Write", located at http://tcd-ie.libguides.com/plagiarism/ready-steady-write.

Signed: ______________________ Date: ______________________

Abstract

This dissertation investigates whether "therapy-speak" related to ADHD and autism has increased in general web discourse over the last decade, and how the surrounding language and framing may have shifted over time. Using Common Crawl as a large-scale web archive, I implement a reproducible pipeline that samples one crawl per year and extracts plaintext (WET) documents for trend estimation and a smaller, deeper corpus for downstream NLP analysis. The study emphasises methodological feasibility: principled sampling, robust filtering, contextual disambiguation, and domain caps to prevent large publishers from dominating the dataset.

Acknowledgements

I would like to thank my supervisor, Dr Tom Paskhalis, for guidance throughout this dissertation, and the MSc in Applied Social Data Science teaching team for methodological training and support.

Contents

List of Figures

List of Tables

1 Introduction

Research Questions

1. How has the prevalence of ADHD- and autism-related terminology in general web discourse changed over approximately the last 10 years?
2. How has the surrounding context and framing of these terms shifted over time (e.g., clinical vs. everyday usage; self-identification; moral/emotional framing)?

2 Literature Review

3 | Materials

3.1 Common Crawl

The primary textual material for this study is derived from Common Crawl, the largest freely available collection of web crawl data (**baackCriticalAnalysisLargest2024**). Common Crawl is distributed in regular crawl releases and is widely used for large-scale NLP corpus construction, though it requires explicit filtering, language identification, and deduplication before analysis (**wenzekCCNetExtractingHigh2019**). Common Crawl is therefore treated as an archival source for public web discourse, not as a representative sample of all online writing or of any population of speakers.

The collection uses one frozen `CC-MAIN` crawl per calendar year from 2014 to 2026 from roughly the same time periods, giving stable cross-sectional snapshots of web text. The pipeline uses WARC Encapsulated Text (WET) files for efficient plaintext scanning and the raw WARC files for document validation and HTML-based main-text extraction (**commoncrawlGetStarted2026**). WET files define the scanned denominator and identify candidate term hits; WARC records are retrieved only for candidate URLs.

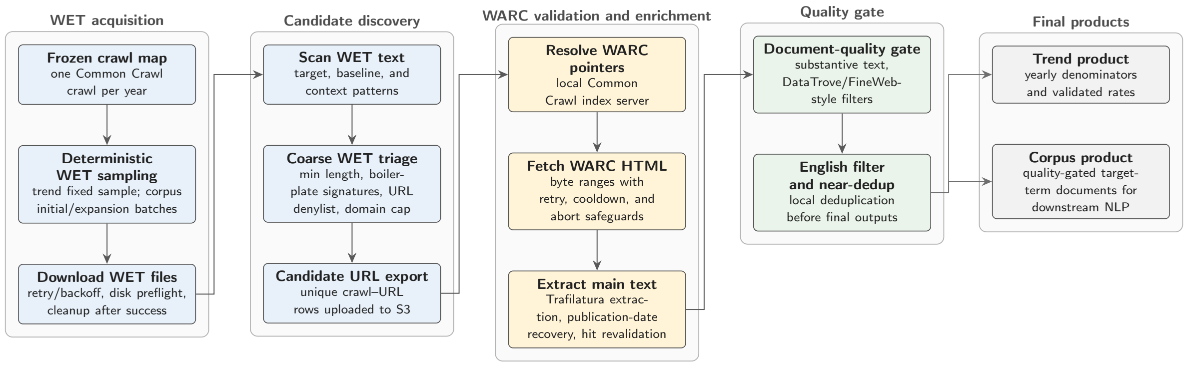

Figure 3.1: Common Crawl collection pipeline used to construct the trend and corpus materials.

Figure 3.1 summarises the material flow. Candidate documents pass through WET-stage triage, WARC pointer resolution, WARC HTML retrieval, main-text extraction, document-quality filtering, English filtering, and local deduplication. The pipeline produces a *Trend Track* for diachronic rate estimates and a *Corpus Track* for downstream semantic and contextual analysis.

3.2 Target Terms

Two target concepts were selected for diachronic lexical semantic change analysis: *ADHD* and *autism*. Target documents were retrieved using the matching expressions shown in Table 3.1. The abbreviation `ASD` was retained only when *autism* occurred within ± 200 characters, reducing false positives from unrelated acronym use.

Table 3.1: Target-term matching expressions.

Concept	Matching expressions
ADHD	`\badhd\b`; `attention[-\s]?deficit`
Autism	`\bautism\b`; `\bautistic\b`; `autism[-\s]?spectrum`; `\bASD\b`

For comparison, I selected three negative, non-clinical emotion terms with sufficient coverage and interpretable usage: *frustration*, *sadness*, and *loneliness*. These terms are not exact semantic controls for ADHD and autism; they provide a baseline for separating target-specific change from broader shifts in negative affective language in Common Crawl.

3.3 NRC–VAD Lexicon

The affective analyses use the NRC Valence, Arousal, and Dominance (VAD) Lexicon v2.1 (**mohammadNRCVADLexicon2025**). The lexicon provides real-valued ratings for approximately 55,000 English words and phrases. Scores range from 0 to 1, where higher values indicate greater valence, arousal, or dominance.

This study uses valence to estimate whether target-term contexts become more positive or negative over time, and arousal to estimate changes in emotional intensity. Dominance is present in the source lexicon but is not used. The ratings were produced through Best–Worst Scaling, which yields fine-grained scores suitable for weighted diachronic collocate analysis.

4 | Method

No-copy–paste outputs

Figures are generated by code into `../reports/figures/stage1_pilot-dev/` and included via `\includegraphics`. Tables are generated by code into `../reports/tables/stage1_pilot-` and included via `\input`.

Pilot validation outputs. Stage 1 pilot validation outputs are provided in the Appendix (see Appendix A), documenting throughput, hit-rate, domain concentration, and the observed boilerplate issue that Stage 1b will mitigate.

5 Results

6 | Discussion

7 | Conclusion

Appendices